BRAC UNIVERSITY

CSE341 - Microprocessors - Summer 2026

Assignment-1

Total Mark: 40

Question 1 [16 marks]

a) Explain the differences between a Microprocessor and a Microcontroller. **[2]**

b) Draw the block diagram of a Microprocessor and a Microcontroller. **[2 + 2 = 4]**

c) Explain the three basic functionalities of a microprocessor. **[3]**

d) Explain what you understand when we say that a microprocessor has a data bus of 16 bits? Explain address bus in detail. **[1 + 1 = 2]**

e) Deduce the maximum size of a memory device compatible with an 8086 microprocessor. Explain the consequences of utilizing an 8GB RAM module with a 32-bit microprocessor. **[1.5 + 1.5]**

f) Explain what an assembler is? Differentiate between RAM and ROM. **[0.5 + 1.5]**

Question 2 [17 marks]

CS = A38Ah, DS = 7F10h, SS = 9C45h, ES = D2B8h, SI = 4E3Ah, DI = 91F0h, SP = B27Ch, BP = 06D2h, IP = 3E2Ah, CX = 0CE7h.

Address	0001Ah	A76CBh	A76CCh	A76CDh	A76CEh
Data	3D	A0	FE	00	0A

TABLE 1

a) Calculate the physical address of the current index in the extra segment. **[2]**

b) I. Can the physical address 0001Ah and A78CBh be segment numbers? Scrutinize with proper logic. **[1 + 1]**

II. Find two valid logical addresses for 0001Ah and A76CBh. **[1.5 + 1.5]**

c) What do you understand by overlapping and non-overlapping segmentation? Write two advantages of segmentation. **[1 + 1 + 2]**

d) Using your answer to **question-(a)**, find the second highest and third lowest logical addresses. **[2+2]**

e) Determine what change will happen in **TABLE 1** after executing the instruction: **[2]**

MOV [IP+0002h],CX.

Question 3 [7 marks]

It is noted that the **physical address = 111h** belongs to multiple different memory segments of the 8086 microprocessor.

a) Calculate the smallest, the largest and the second largest segment numbers and the corresponding logical addresses. Failure to provide proper logical explanation will not be entertained. **[1.5 + 1.5 + 1.5]**

b) Determine the fourth largest Physical address possible for the largest segment number you found above. **[1]**

c) Assuming you are using non-overlapping segmentation, calculate the segment that the Physical address = 88292h belongs to and find the logical address. **[1.5]**